

Wireshark Traffic Analysis Report

Fake Software Download Infection

Case Source	Malware-Traffic-Analysis.net - 2025-01-22 Traffic Analysis Exercise: Download from Fake Software Site
Analyst	Nikola Jankovic
Report Type	Network Traffic Analysis / SOC Portfolio Report
Primary Evidence	PCAP traffic, DNS/HTTP/TLS indicators, suspicious IP/domain activity, IOC summary
Severity	High

Title:

Wireshark Traffic Analysis - Fake Google Authenticator Software Download Infection

Scenario:

A network traffic investigation was performed using the Malware-Traffic-Analysis.net PCAP exercise titled "2025-01-22 - Traffic Analysis Exercise: Download from Fake Software Site." The scenario describes a SOC analyst receiving a report that a coworker downloaded a suspicious file after searching for Google Authenticator. A packet capture was retrieved and reviewed to confirm whether an infection occurred.

The investigation focused on identifying the infected Windows client, hostname, user account, likely fake Google Authenticator domain, suspicious external IP addresses, and possible command-and-control communication patterns. This report is structured as a GitHub-ready cybersecurity portfolio project demonstrating practical network traffic analysis skills.

Tools used:

- Malware-Traffic-Analysis.net PCAP file
- Wireshark
- tcpdump concepts / packet-filtering methodology
- DNS traffic analysis
- HTTP request analysis
- TLS/SNI inspection
- Wireshark Conversations and I/O Graphs
- MITRE ATT&CK; Framework
- Text editor for IOC documentation
- GitHub for portfolio documentation

Alert / initial evidence:

The initial evidence was a PCAP file from the Malware-Traffic-Analysis.net case. The case background states that a user downloaded a suspicious file after searching for Google Authenticator and that a packet capture was retrieved for confirmation of the infection.

LAN segment information from the PCAP:

- LAN segment range: 10.1.17[.]0/24
- Domain: bluemoontuesday[.]com
- Domain controller: 10.1.17[.]2 - WIN-GSH54QLW48D
- AD environment: BLUEMOONTUESDAY
- Gateway: 10.1.17[.]1
- Broadcast address: 10.1.17[.]255

Investigation steps:

1. Opened the PCAP file in Wireshark and confirmed packets were loaded correctly.
2. Reviewed the network environment and identified the LAN range and domain context.
3. Filtered for the infected host IP address to review internal and external communication.
4. Used DHCP traffic to identify the infected host name.
5. Used Kerberos traffic to identify the user account associated with the infected Windows client.

6. Reviewed DNS queries to identify fake software and suspicious domains.
7. Reviewed TLS/SNI traffic to identify suspicious HTTPS destinations.
8. Reviewed HTTP traffic for suspicious web requests or download activity.
9. Filtered for suspicious external IP addresses and reviewed related packet activity.
10. Used Follow TCP Stream to inspect suspicious communication flows.
11. Used Statistics > Conversations to review external connections and possible C2 destinations.
12. Used Statistics > I/O Graphs to observe repeated activity that may indicate beaconing.
13. Extracted indicators of compromise and documented them in an IOC file.
14. Mapped the observed behaviour to MITRE ATT&CK; techniques.
15. Documented impact, severity, and recommended defensive actions.

Findings:

Infected host evidence:

Traffic analysis identified the infected Windows client as 10.1.17.215. Filtering on this IP showed traffic involving the suspected host and external destinations.

```
Wireshark filter:  
ip.addr == 10.1.17.215
```

Host and user attribution:

DHCP and Kerberos traffic were used to link the activity to a specific Windows system and user account.

- Infected IP address: 10.1.17.215
- Hostname: DESKTOP-L8C5GSJ
- Username: shutchenson

```
DHCP filter:  
dhcp  
  
Kerberos filter:  
kerberos
```

DNS / suspicious domain evidence:

DNS analysis identified fake software-related domain activity connected to the Google Authenticator lure. The domain names indicate likely impersonation of legitimate software download infrastructure.

- Suspicious domain: authenticatoor[.]org
- Related fake software domain: google-authenticator.burleson-appliance[.]net

```
DNS filter:  
dns
```

TLS/SNI indicators:

TLS/SNI analysis was used to inspect HTTPS server names visible in Client Hello packets. This helps reveal destination domains even when web content is encrypted.

```
TLS/SNI filters:  
tls.handshake.extensions_server_name  
ssl.handshake.extensions_server_name
```

HTTP traffic indicators:

HTTP request analysis was used to look for web requests, download paths, host headers, and suspicious content related to the fake software site.

```
HTTP request filter:  
http.request
```

Suspicious IP / possible C2 evidence:

Traffic analysis identified communication involving suspicious external infrastructure. The external IP 5.252.153.241 should be treated as a possible command-and-control or malicious infrastructure indicator within this portfolio case.

```
Suspicious IP filter:  
ip.addr == 5.252.153.241
```

Possible C2 / beaconing pattern:

Wireshark Conversations and I/O Graphs should be reviewed to identify repeated outbound communication from the infected host to suspicious external destinations. Repeated or periodic communication may support a possible C2/beaconing assessment.

Indicators of Compromise:

Indicator Type	Value
Infected IP Address	10.1.17.215
Hostname	DESKTOP-L8C5GSJ
Username	shutchenson
Suspicious Domain	authenticatoor[.]org
Related Domain	google-authenticator.burleson-appliance[.]net
Suspicious IP / Possible C2	5.252.153.241
LAN Domain	bluemoontuesday[.]com
AD Environment	BLUEMOONTUESDAY

MITRE ATT&CK mapping:

Technique	ID	Relevance
User Execution	T1204	User likely downloaded or interacted with suspicious software content.
Phishing / Social Engineering	T1566 / related	The lure used fake software download infrastructure impersonating Google Authenticator.
Ingress Tool Transfer	T1105	Suspicious content may have been retrieved from external infrastructure.
Application Layer Protocol	T1071	Suspicious communication occurred over web/TLS-related protocols.
Command and Control	T1071 / related	Repeated external communication may indicate C2-style activity.
Obfuscated Files or Information	T1027	Relevant if encoded or suspicious content is observed in TCP streams or downloaded content.

Impact assessment:

The traffic indicates that a Windows host accessed fake software-related infrastructure and communicated with suspicious external destinations. The activity is consistent with a fake software download infection scenario. The likely business impact includes endpoint compromise risk, possible malware download/execution, potential credential theft, possible command-and-control communication, and the need for endpoint containment and forensic review.

Severity is assessed as High because the case involves a named infected host, suspicious domain activity, and communication with potentially malicious external infrastructure.

Recommended actions:

- Isolate the affected host 10.1.17.215 from the network.
- Investigate the endpoint for downloaded files, suspicious processes, persistence, and browser history.
- Block authenticator[.]org and google-authenticator.burleson-appliance[.]net at DNS/proxy/security controls.
- Block or monitor traffic to 5.252.153.241.
- Reset credentials for the affected user account shutchenson if compromise is suspected.
- Run a full EDR/AV scan on the endpoint.
- Review proxy, DNS, firewall, and authentication logs for related activity.
- Create SIEM alerts for suspicious fake software download domains and unusual outbound connections.
- Educate users to download software only from official vendor websites.

Conclusion:

The Wireshark investigation confirmed suspicious network activity consistent with a fake software download infection. The infected Windows host was identified as 10.1.17.215, with hostname DESKTOP-L8C5GSJ and user account shutchenson. DNS, TLS/SNI, HTTP, and IP-based analysis identified suspicious fake Google Authenticator-related domains and a suspicious external IP address.

This project demonstrates practical SOC Analyst skills including PCAP analysis, infected host identification, DNS/HTTP/TLS investigation, TCP stream review, possible C2/beaconing analysis, IOC extraction, MITRE ATT&CK; mapping, and defensive recommendation writing.

Screenshot evidence checklist:

- 01_pcap_loaded_wireshark.png
- 02_infected_host_ip_10_1_17_215.png
- 03_dhcp_hostname_desktop_l8c5gsj.png
- 04_kerberos_username_shutchenson.png
- 05_dns_fake_authenticator_domain.png
- 06_tls_sni_suspicious_domain.png
- 07_http_request_suspicious_download.png
- 08_suspicious_ip_5_252_153_241.png
- 09_follow_tcp_stream_suspicious_content.png
- 10_tcp_conversations_c2_pattern.png
- 11_io_graph_possible_beaconing.png
- 12_ioc_summary.png

GitHub repository structure:

```
wireshark-fake-software-c2-analysis/
|-- README.md
|-- Wireshark_Traffic_Analysis_Report.pdf
|-- screenshots/
| |-- 01_pcap_loaded_wireshark.png
| |-- 02_infected_host_ip_10_1_17_215.png
| |-- 03_dhcp_hostname_desktop_l8c5gsj.png
| |-- 04_kerberos_username_shutchenson.png
| |-- 05_dns_fake_authenticator_domain.png
| |-- 06_tls_sni_suspicious_domain.png
| |-- 07_http_request_suspicious_download.png
| |-- 08_suspicious_ip_5_252_153_241.png
| |-- 09_follow_tcp_stream_suspicious_content.png
| |-- 10_tcp_conversations_c2_pattern.png
```

```
| |-- 11_io_graph_possible_beaconing.png  
| `-- 12_ioc_summary.png  
|-- iocs/  
`-- iocs.txt
```

Sources:

Report structure modeled after the attached Basic Malware Triage Report. Case evidence based on Malware-Traffic-Analysis.net:
2025-01-22 - Traffic Analysis Exercise: Download from Fake Software Site.